$$CV_K = \frac{1}{n}\sum_i \left(y_i - \hat{f}^{(-k(i))}(x_i)\right)^2 = \frac{1}{k}\sum_{k=1}^{K} MSE_k$$

1.a) What does k-fold cross-validation estimate?
the expected error of ~~the~~ specific combination of hyperparameters

b) The K numbers are not ~~independent~~ since there ~~are~~ overlaps ~~bob~~ with data in the folds.
As K increases, the bias decreases but the usefulness of each fold also decreases since they are less independent. SJ_K is optimistic when variance is large.

c) Scaling. The scaler must be fitted inside each fold when the input features have different scales e.g. feature 1 = ~~height~~ cm, 2 = kg, to have a mean of 0 and s.d of 1, so that there isn't a separate mean and s.d ~~for~~ the validation options.
- Pipeline helps to connect ~~prep~~ the steps to ensure there is no leakage

d) "the best learning rate is 0.01, with a test MSE of 0.0094"
- the learning rate shouldn't be chosen on the test set.
↳ no cross validation, the test set could be unrepresentative of the data
- no comparison of other MSE values,

e) The number of epochs is a hyperparameter because it ~~cannot~~ has to be selected and kept the same before the parameters are chosen. It cannot

be chosen by the training error, which is non-increasing in the epochs. Select a max epoch number where overfitting will occur, and then average the validation loss/accuracy for each epoch across folds to find the global minimum/maximum.

f) The one-standard error rule:

$$\lambda_{1SE} = \max \{ \lambda : CV(\lambda) \leq CV(\lambda_{min}) + SE(\lambda_{min}) \}$$

it finds the largest penalty. So you look at the error of the best model and add one s.e to it, and any model with error below this line is considered statistically equivilent.